

PROJECT APPRAISAL

Module 5

Introduction to Economic Appraisal



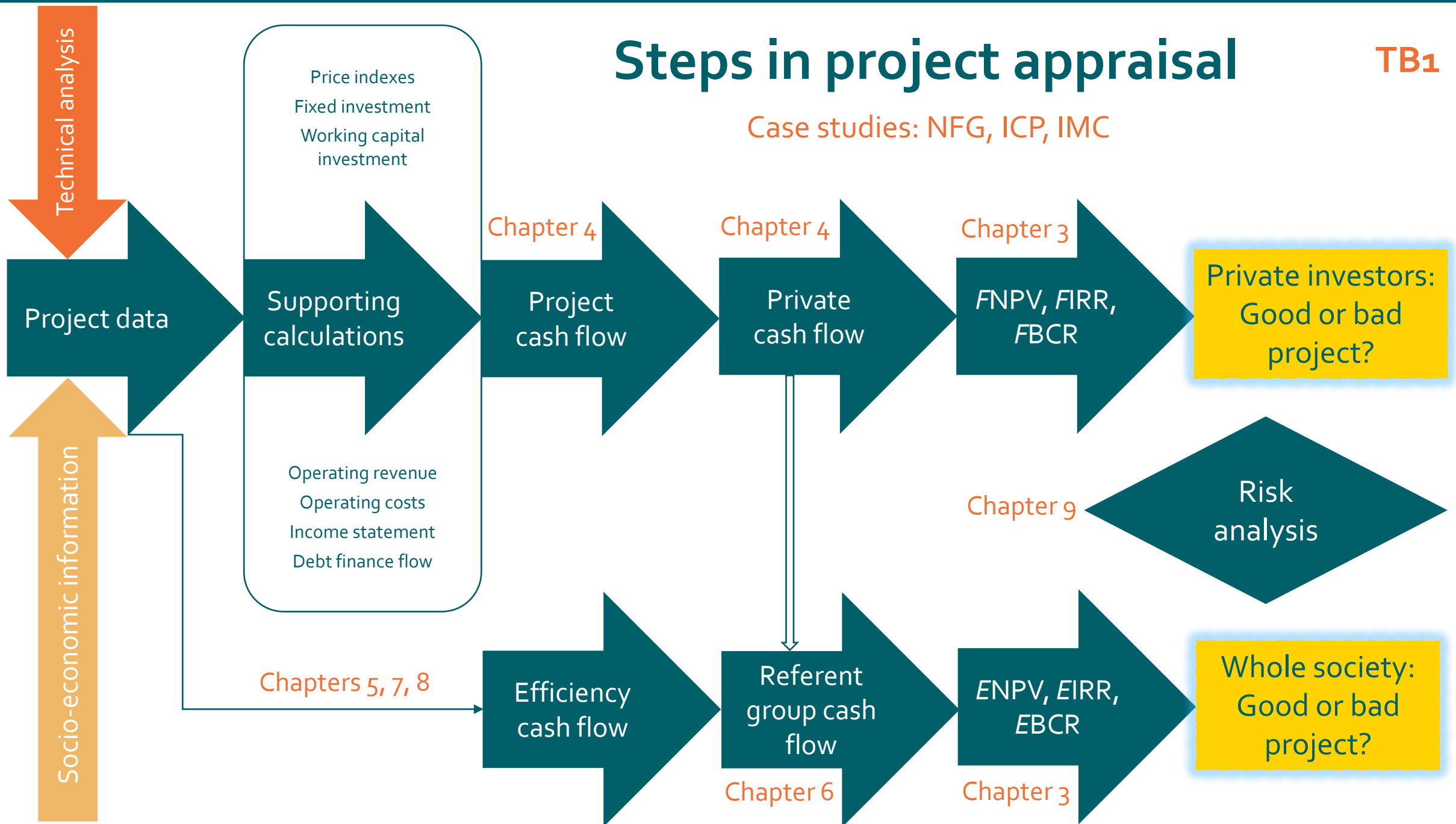
Outline – Module 5

- The rationale of efficiency analysis
- Cost-benefit analysis model (a review)
- Opportunity cost of a project's inputs
- Economic value of a project's outputs
- Shadow prices: An introduction
- From market analysis to efficiency analysis
- Some illustrative examples

Steps in project appraisal

TB1

Case studies: NFG, ICP, IMC



Efficiency analysis

An introduction



The rationale of efficiency analysis

- In financial analysis, we valued the project's outputs and inputs at **market prices** to calculate financial cash flows (FCFs). However, the FCFs are an incomplete measure of the effect of the project for two reasons:
 - The market price of a project's input or output may not accurately measure its **social benefit or cost**;
 - Some inputs or outputs have **no market prices**, and are valued at zero in financial analysis.

The rationale of efficiency analysis

- All real-world markets may be **non-competitive** and **distorted**, and markets in some critical goods and services do not exist.
- Even in a competitive and undistorted market, a **large-scale** (public) project may cause a change in the price of an input or an output.
- Therefore, the financial analysis will be an imperfect judge of whether a project contributes to an improvement in the efficiency of resource allocation.

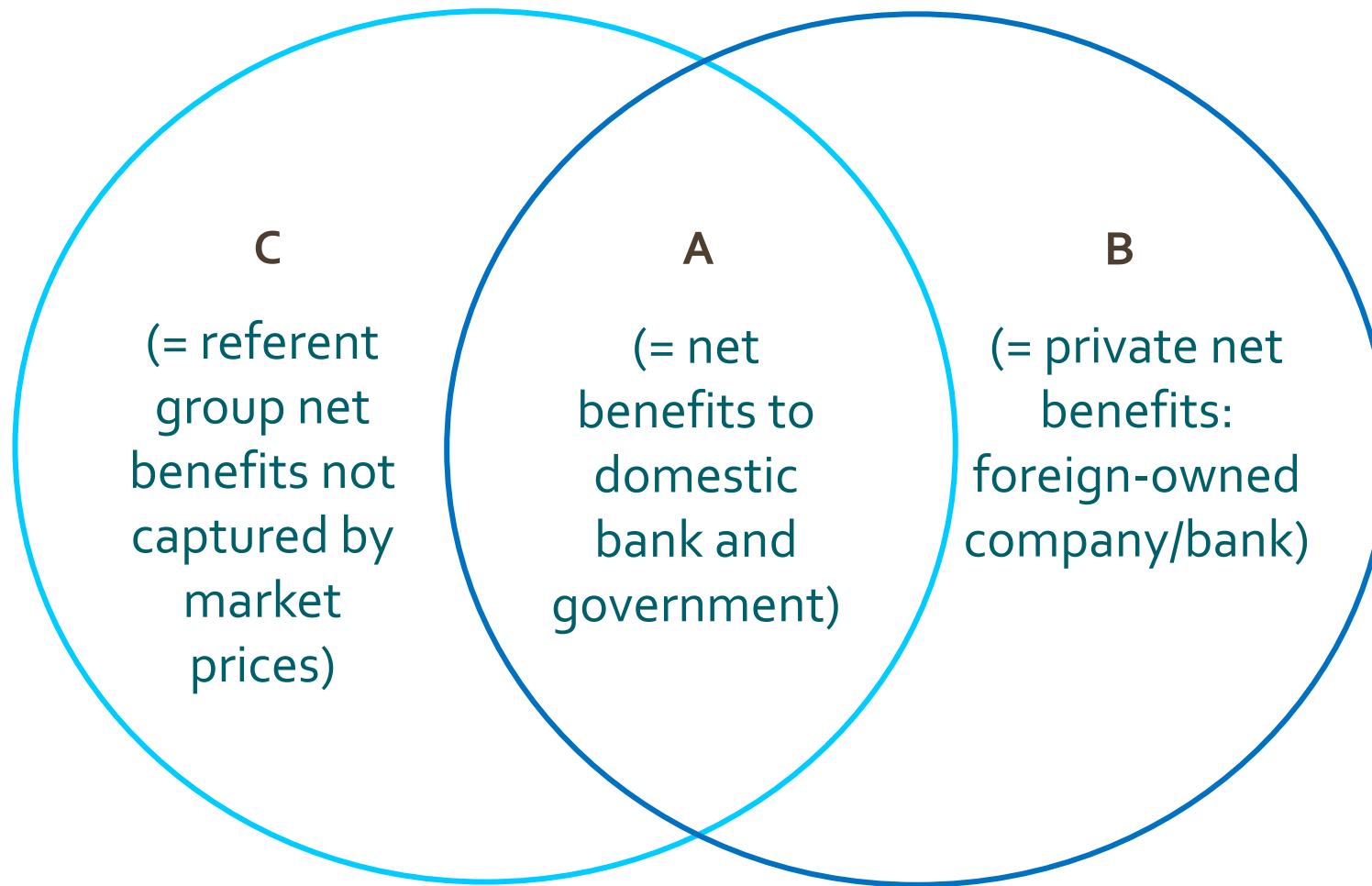
The rationale of efficiency analysis

- We solve these problems by developing **shadow prices** for valuing the project's outputs and inputs:
 - Adjust market prices to provide a measure of benefit or cost to the economy;
 - Identify and measure the values of non-marketed goods and services of a project.
- These are not prices that are actually paid to, or received by any economic agent; they are purely to account for effects that are not measured by market prices.

Efficiency analysis

The cost-benefit analysis model (Module 1)





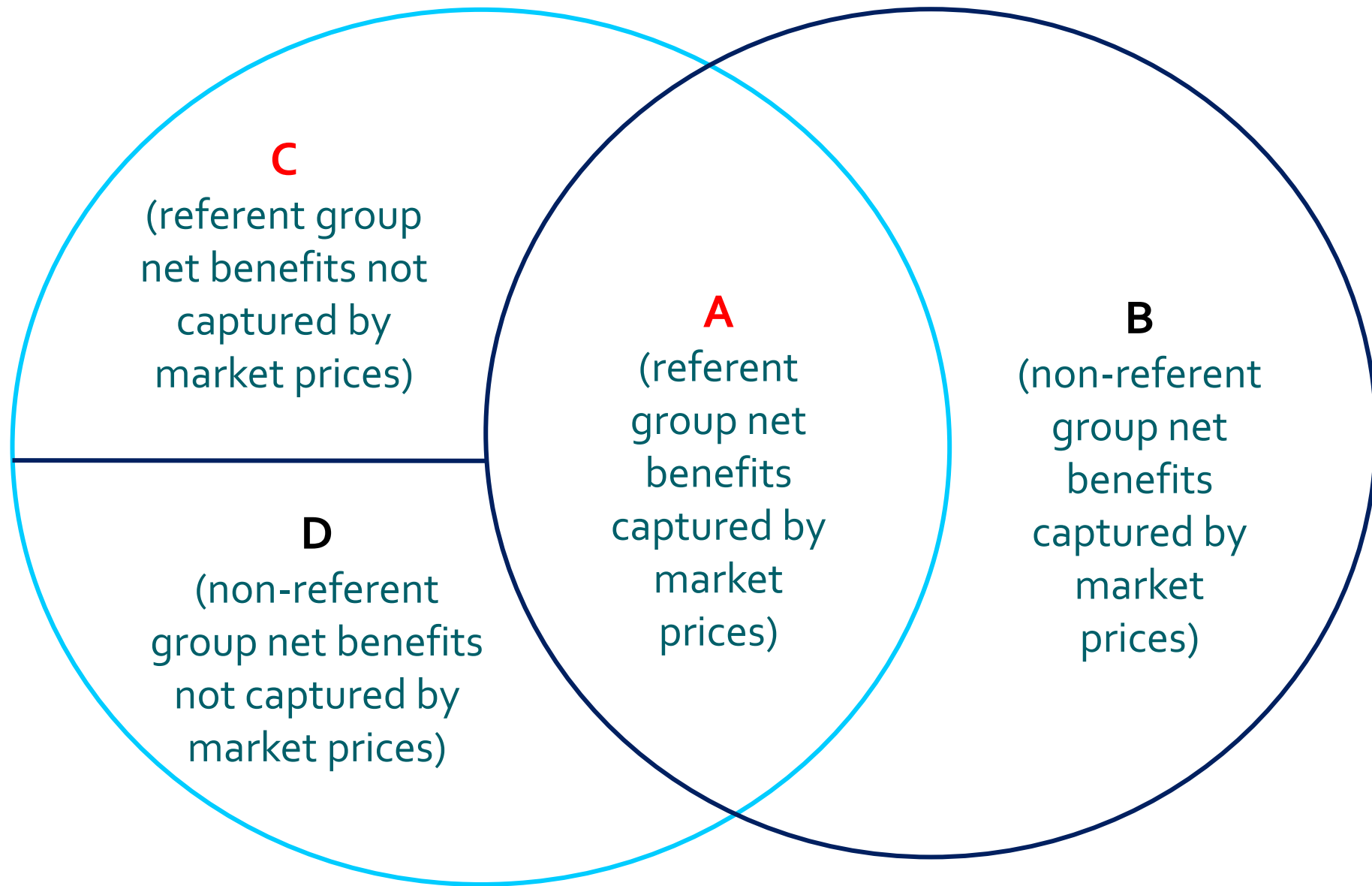
A: Referent Group
(market prices)

B: **Private Analysis**
(market prices)

C: Referent Group
(non-market
prices)

A + B + C: **Efficiency/Economic Analysis** (market and
non-market prices)

A+B:
**Market/Project
Analysis** (market
prices)



A + C: Referent Group Analysis

The cost-benefit analysis model

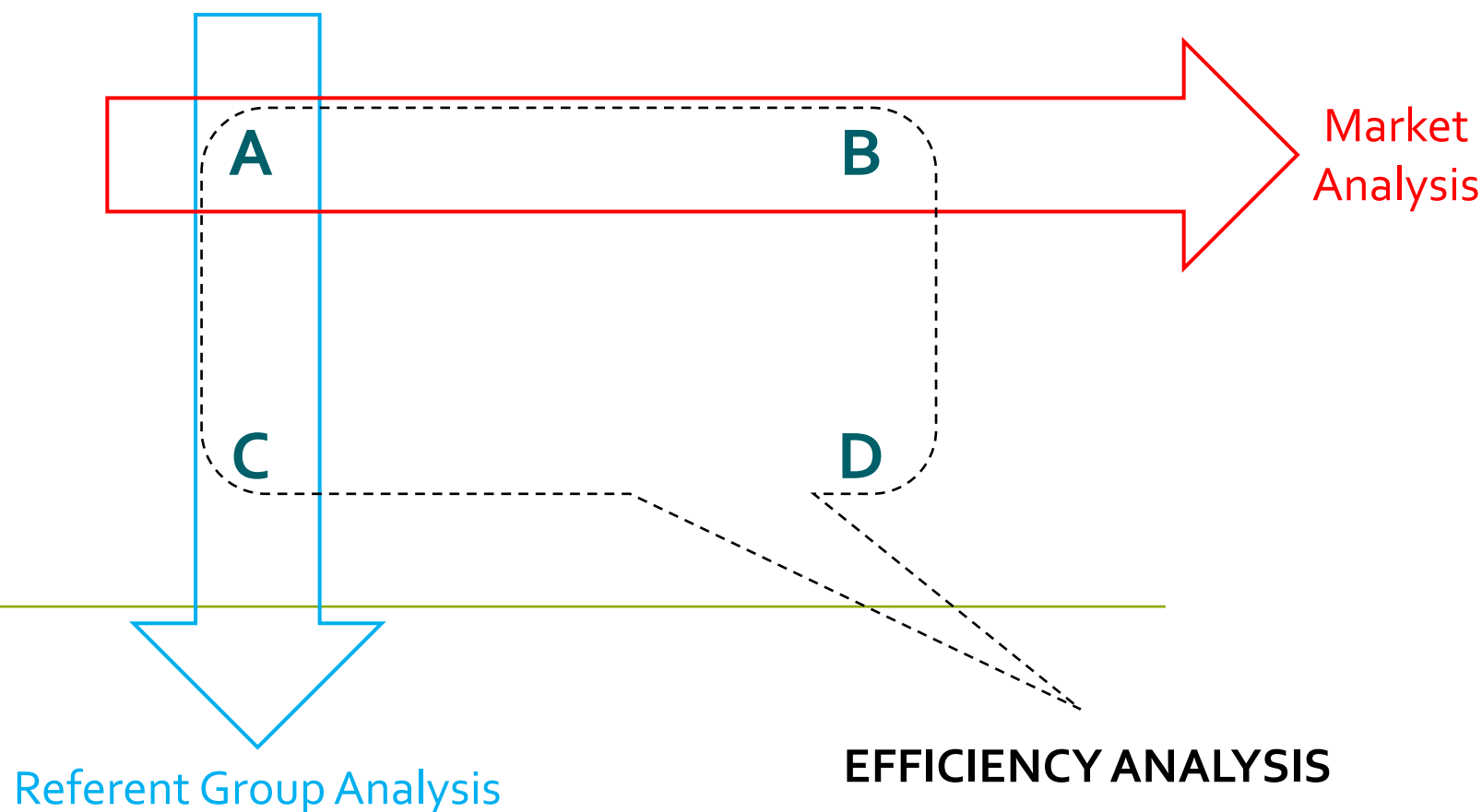
The project net benefits

Referent group

Non-referent group

Net benefits are measured by market prices

Net benefits are not measured by market prices



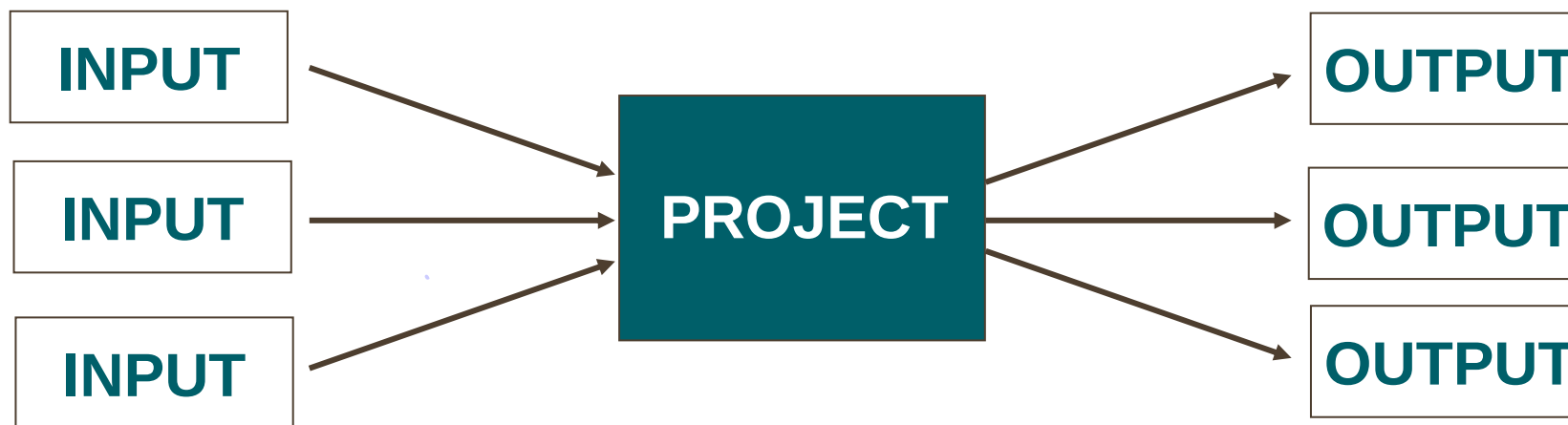
Economic value

Opportunity cost of a project's inputs



Opportunity cost

Costs and benefits of a project



**What are the social costs
of these inputs?**

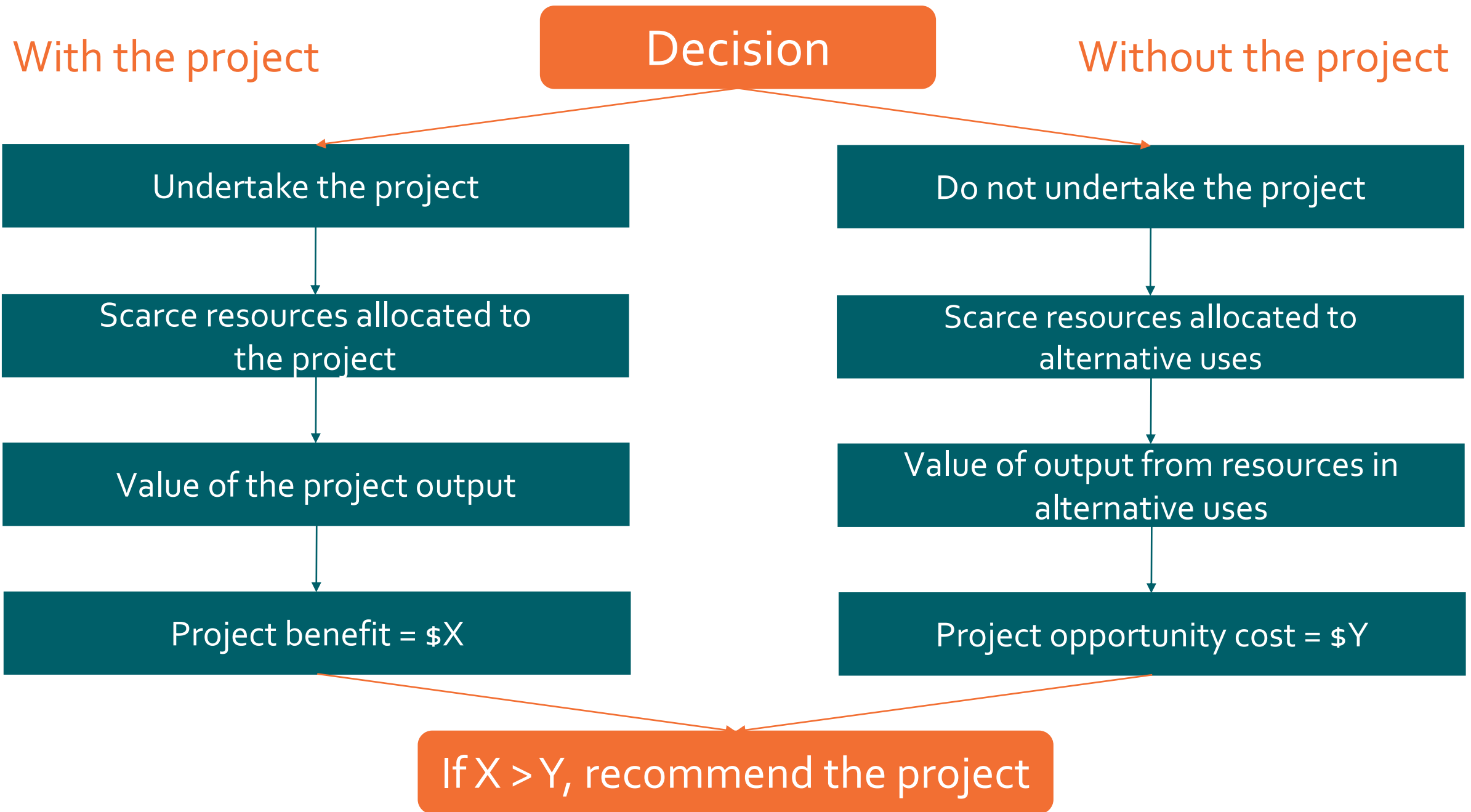
**What are the social benefits
of these outputs?**

Opportunity cost

Economic cost of an input used in a project is measured by the economic value of this input in its best alternative use.

In other words, opportunity cost measures the value of goods and services which would have been produced by the inputs had they not been used to undertake the project.

Normally, market price of an input will be a good measure of opportunity cost. However, in some cases, this is no longer correct. Why?



Opportunity cost

- Consider an input is subject to an *valorem* indirect tax, and the output is also subject to such a tax. The profit-maximizing firm will employ the input up to the point at which:

$$P_{QS} * MPP_X = P_X(1 + t_X)$$

- P_{QS} is the price suppliers of the output receive
- MPP_X is the marginal physical product of the input (the amount of extra output resulting from adding one extra unit of the input)
- P_X is the price suppliers of the input receive
- t_X is the rate of indirect tax on the input.

Opportunity cost

- Now suppose that there is also a tax on the firm's output so that:

$$P_{QB} = P_{QS}(1 + t_Q)$$

- P_{QB} is the price buyers pay for the product produced by the firm
- t_Q is the rate of indirect tax on that product.
- Thus, the opportunity cost of diverting a unit of the input from use of the firm:

$$OC_X = P_{QB} * MPP_X = P_X(1 + t_X)(1 + t_Q)$$

(the value buyers of the firm's product would have placed on the output foregone as a result of diverting a unit of the input to another use)

Opportunity cost

Consider a project that recruits employees as an input.

What is the financial cost of this labour?

Financial cost of using labour is simply the price paid to the employees. This price is often measured by the market wage.

Opportunity cost

What is the economic cost of this labour?

Economic cost of using this labour in the same project is measured by the opportunity cost of labour; that is the value of labour in the best alternative use.

Economic cost depends on whether the employee is being employed or unemployed.

If the employee is being employed elsewhere, the opportunity cost can be measured by the market wage of the replaced job. In this circumstance, economic cost of labour is equal to financial cost of labour. We often refer this as skilled labour.

However, if the employee is being unemployed, the opportunity cost of labour may be very low. In such a case, economic cost of labour is smaller than financial cost of labour. We often refer this as unskilled labour.

Other examples of opportunity cost will be discussed in Module 6!

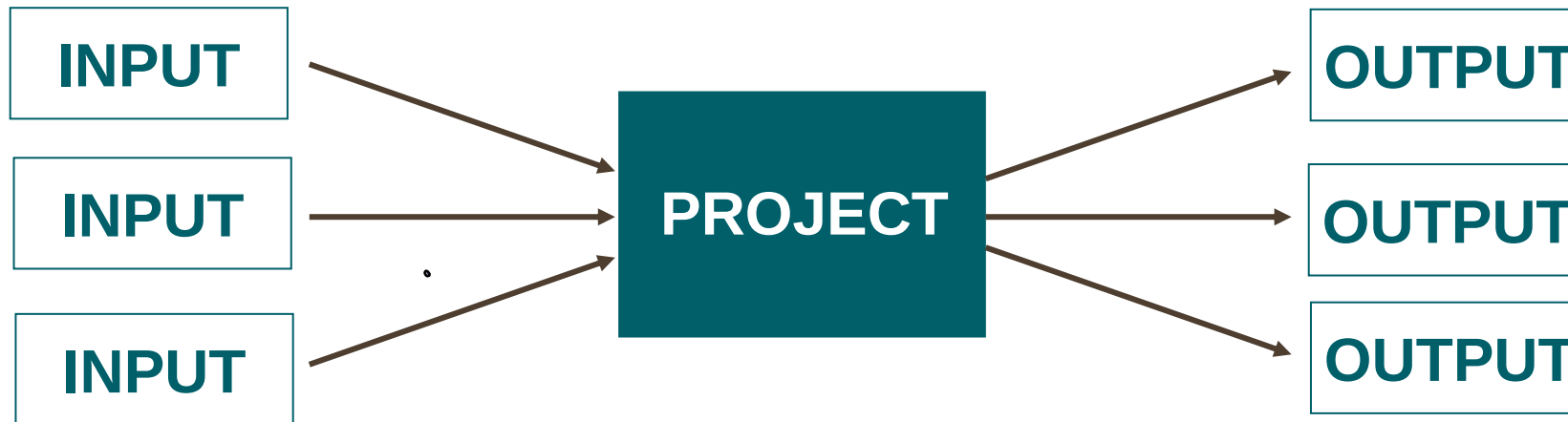
Economic value

Economic benefit of a project's outputs



Economic benefit

Costs and benefits of a project



What is the social costs of these inputs?

What is the social benefits of these outputs?

Consider a project that increases the street trees ...

- Suppose an individual with an initial welfare is W_0 from monetary income level of Y_0 and enjoying the street trees of T_0 :

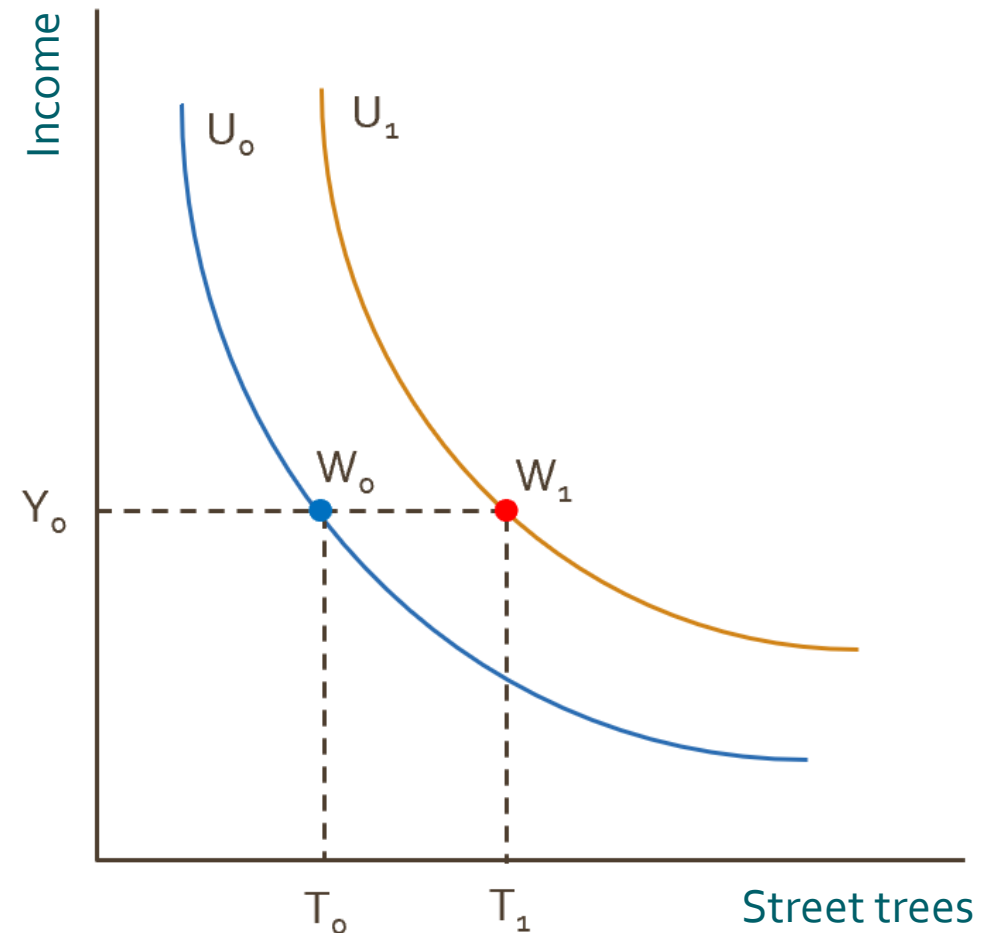
$$W_0(Y_0, T_0)$$

- This individual is offered a project that increases street trees to T_1 . More street trees will increase his welfare to W_1 :

$$W_1(Y_0, T_1)$$

- Because this individual welfare increases with the project, we expect:

$$W_1(Y_0, T_1) > W_0(Y_0, T_0)$$

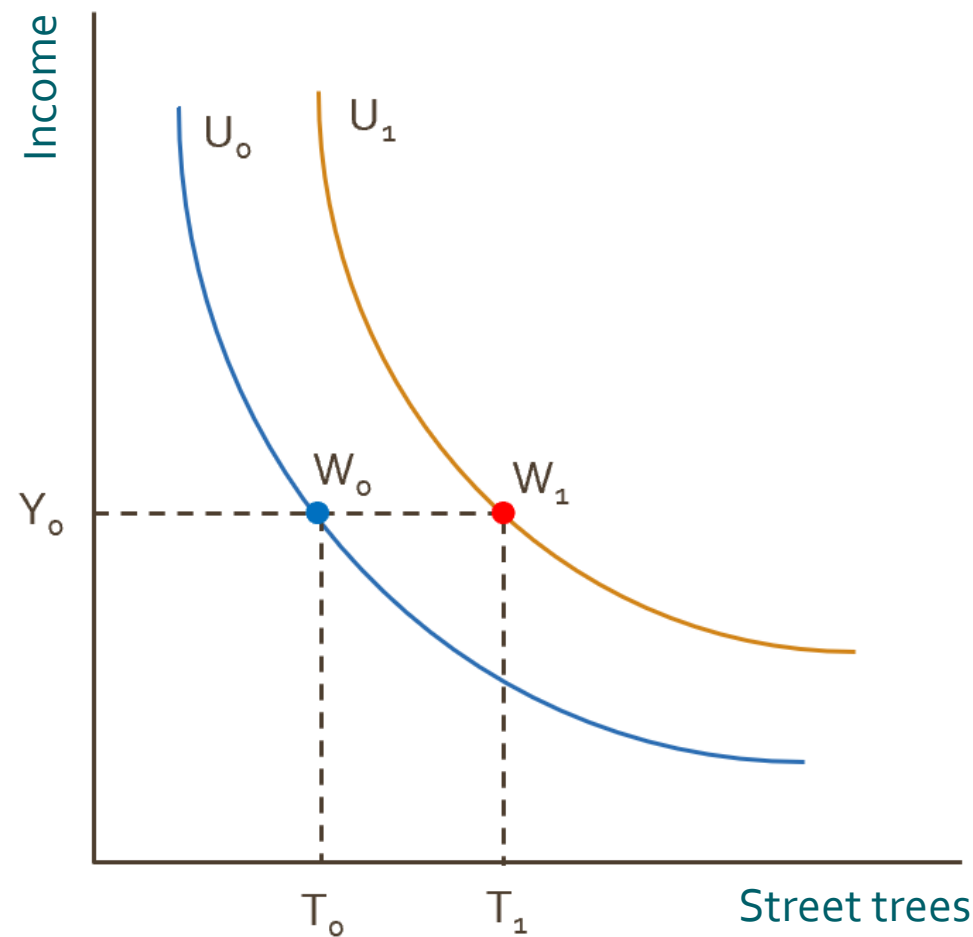


- To evaluate the relevance of the project, and compare its costs and benefits, we need to know the value of welfare increase with the project; that is, how much is the difference between W_1 and W_0 ?

$$\Delta W = W_1(Y_0, T_1) - W_0(Y_0, T_0)$$

- How will we measure this change in welfare? That is, how large is the difference between W_1 and W_0 with the project?

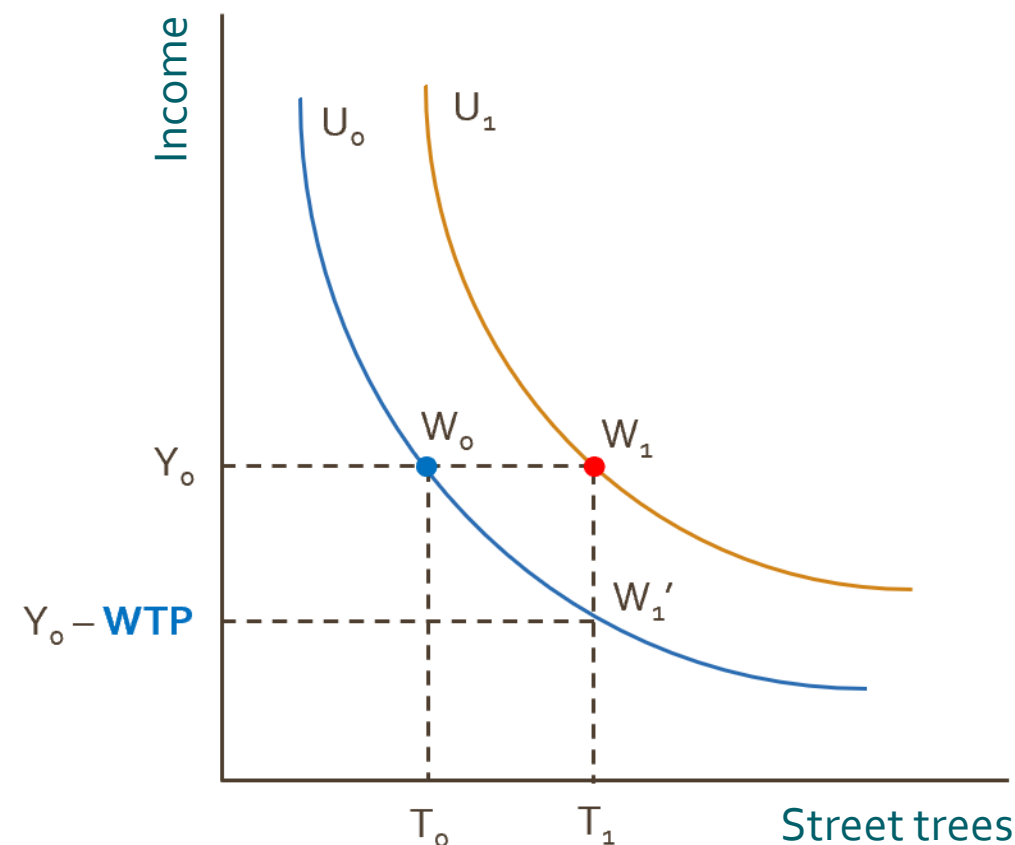
There are two different approaches



Approach 1: WTP measure

- Determine the **maximum amount of money** that an individual is willing to pay for the proposed project.
- An individual is asked to make choice between the two combinations between income and street trees that give him the same utility level (U_0):
 - The first combination which reduces his income but increases the street trees; and
 - The second combination which both income and street trees remain unchanged.

$$W_0(Y_0, T_0) = W'_1(Y_0 - \text{WTP}, T_1)$$

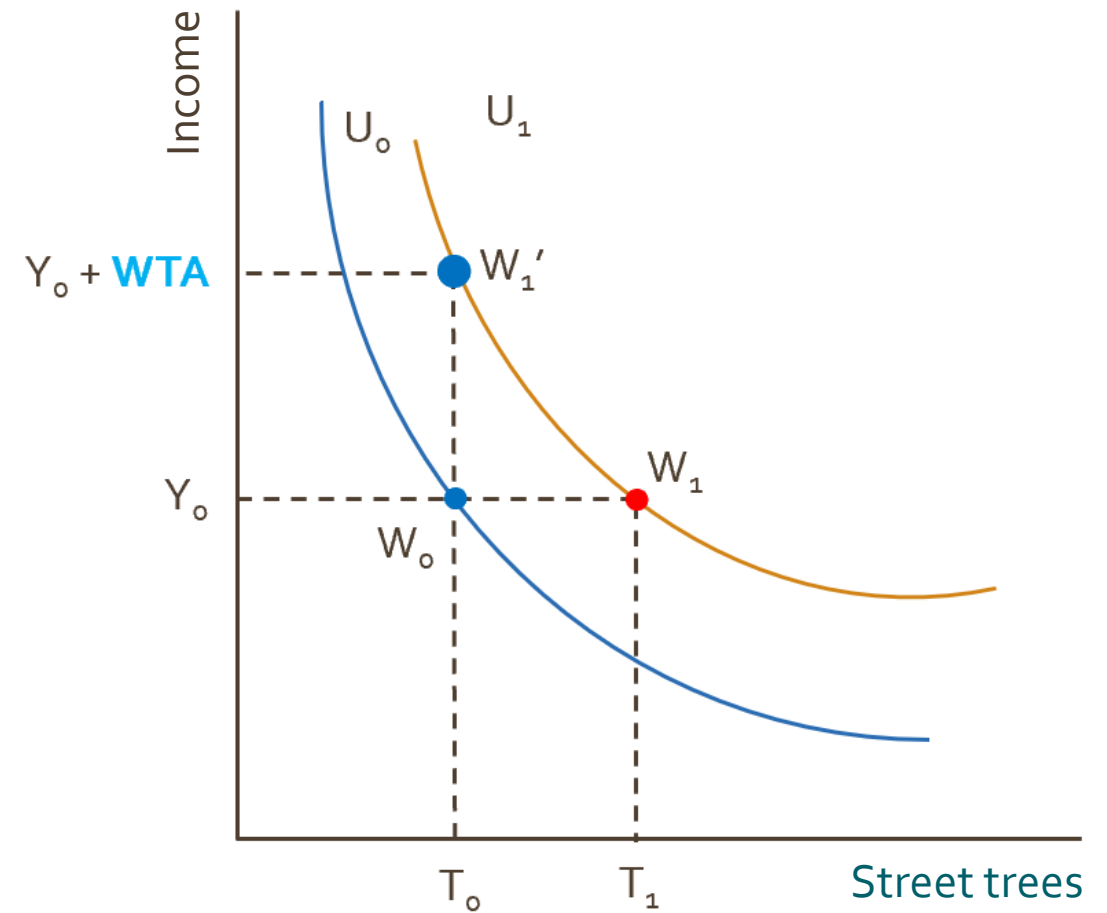


- WTP is defined as the amount of money that makes two combinations of income and street trees give him the same welfare level. This is the maximum amount of money that he is willing to pay for a positive change in welfare from the increased street trees.
- The **maximum WTP** is a measure of economic value for a change in welfare due to an increase in street trees from T_0 to T_1 .

Approach 2: WTA measure

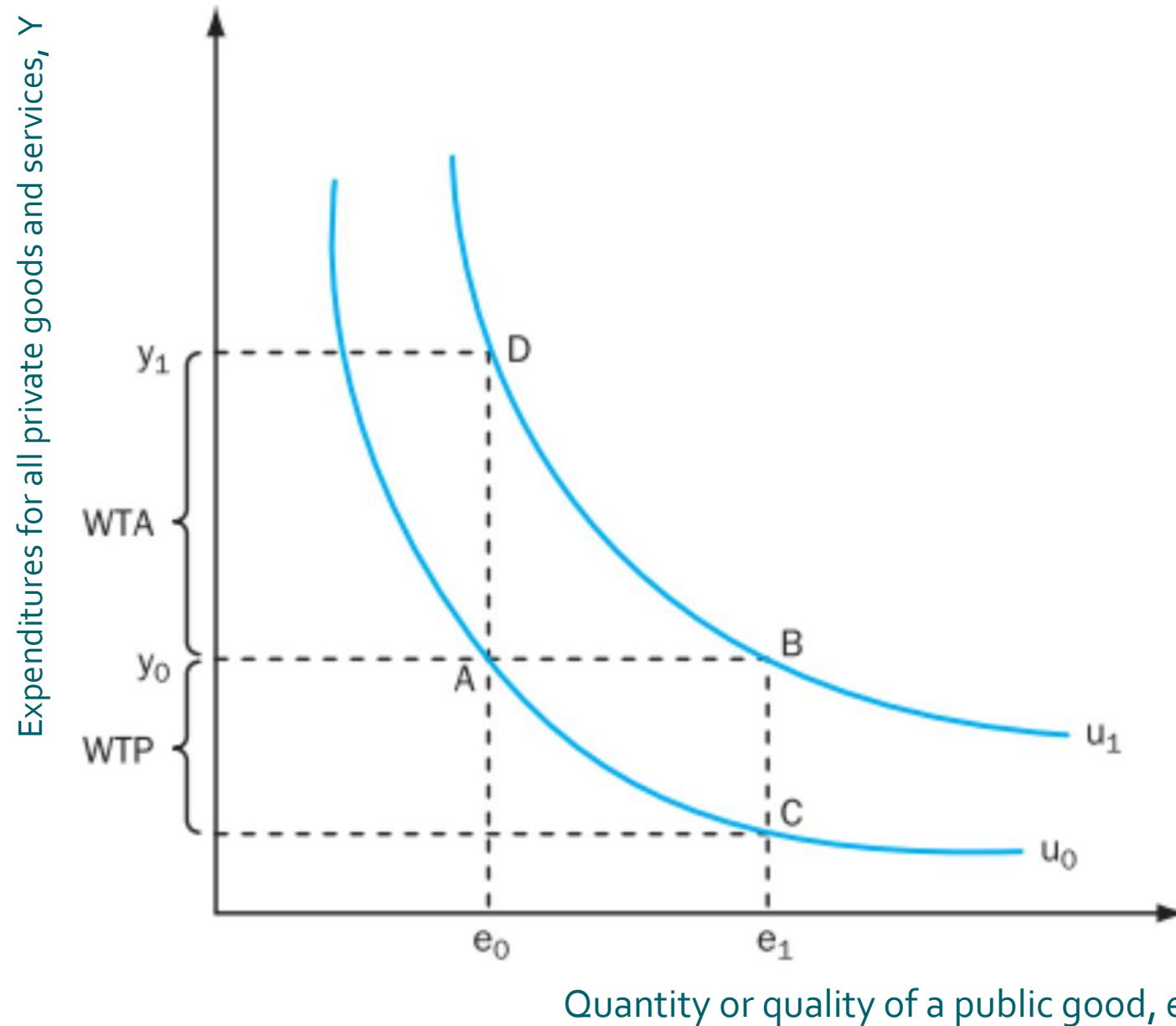
- Determine the **minimum amount of money** that an individual is willing to accept a compensation to give up the proposed project.
- An individual is asked to make choice between the two combinations between income and street trees that give him the same utility level (U_1):
 - The first combination which income is unchanged but the street trees are increased; and
 - The second combination which income increases but street trees remain unchanged.

$$W_1(Y_0, T_1) = W'_1(Y_0 + WTA, T_0)$$



- WTA is defined as the amount of money that makes two combinations of income and street trees give him the same welfare level. This is the minimum amount of money that he is willing to accept a compensation for a positive change in welfare from the increased street trees.
- The **minimum WTA** is a measure of economic value for a change in welfare due to an increase in street trees from T_0 to T_1 .

A brief summary



Perman et al., 2011 (414).

Shadow prices

How are they different from market prices?



Shadow prices

For each project's input:

- What is the social cost of using this input to undertake the project?
- In some cases, the market price that a project pays to supplier to possess and use an input provides us a correct measure of that input in economic appraisal (i.e. economic cost equals financial cost).

Economic costs

But:

- In some cases, financial cost (measured by market prices) will not be a good measure of economic cost. For example, there may be some kind of market distortions (such as taxes, subsidies, import duties, export subsidy, etc).
- In other cases, there may be no market, and thus no market price for an input of the project (for example, a project uses environment for its discharge of waste water).

In such cases, we need to either adjust market prices or estimate shadow prices for the inputs in economic appraisal.

Economic benefits

For each project's output:

- What is the social benefit of providing this output to society?
- In some cases, the market price that a project receives from selling its output provides us a correct measure of that output in economic appraisal (i.e. economic benefit equals financial benefit).

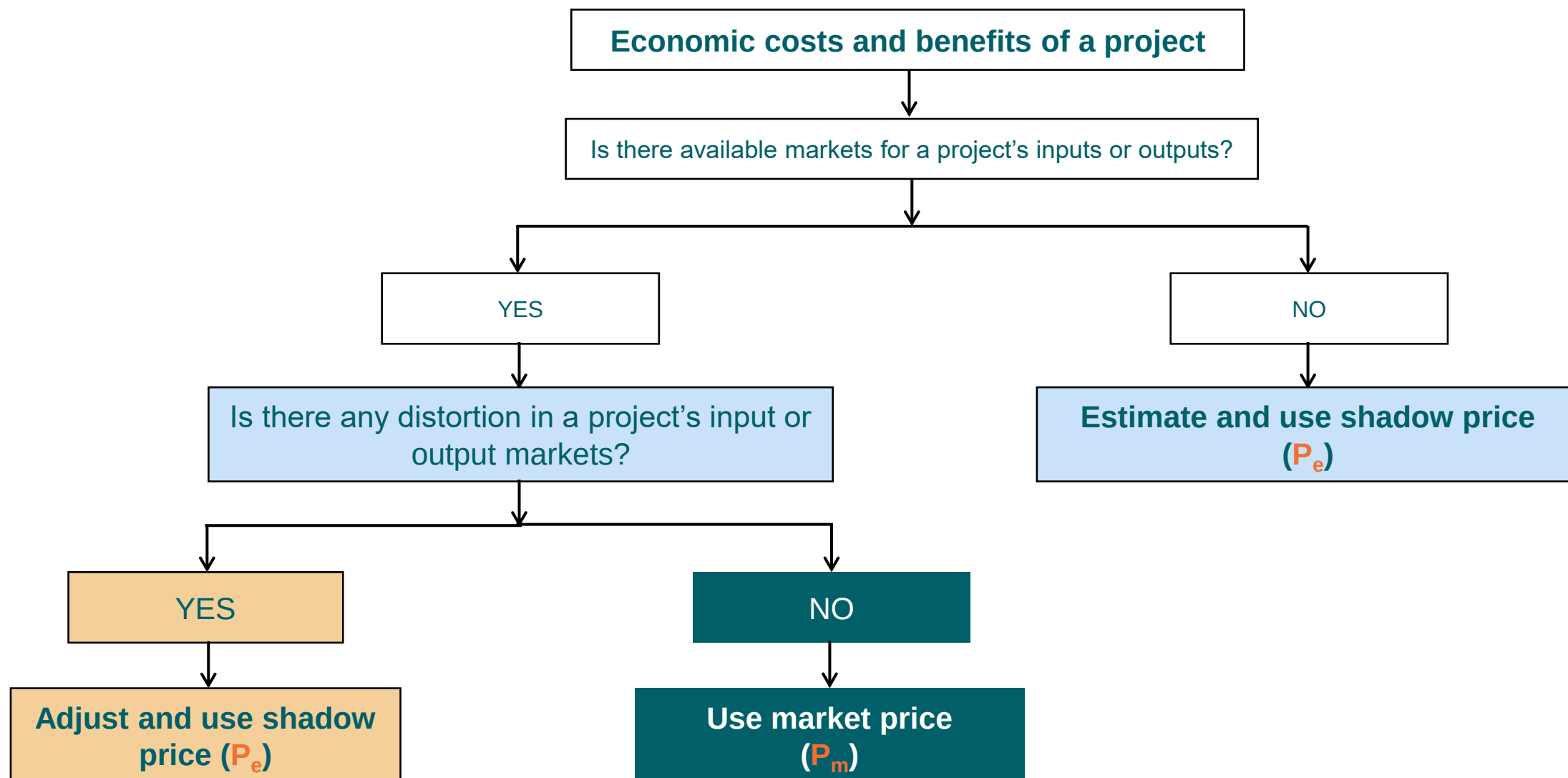
Economic benefits

But:

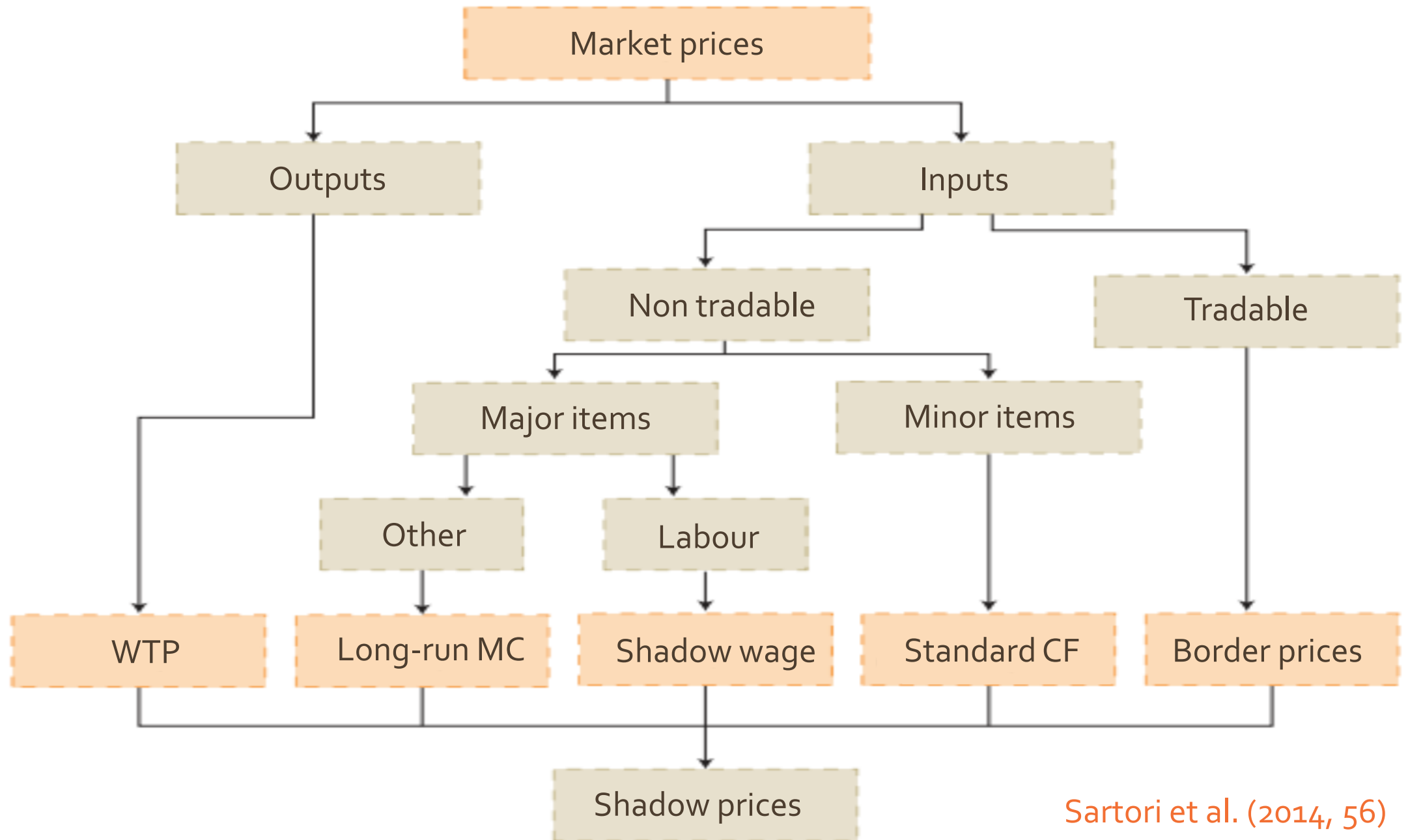
- In some cases, market prices are not good measures of economic benefit due to some kinds of market distortions such as price control.
- In other cases, there may be no market, and thus market price for goods and services produced by a project (for examples, travel time saving from less traffic, natural conservation).

In such cases, we need to either adjust market prices or estimate shadow prices for the outputs in economic appraisal.

Economic benefits



From market prices to shadow prices



Sartori et al. (2014, 56)

From market analysis

To efficiency analysis of a project



1) Adjustments of market prices

For inputs or outputs used or produced by a project that are traded and have market prices, the analyst needs to convert market prices to efficiency prices and enters them together in the project data.

Important note: In the project data, the relationship between market price and efficiency price of an item must be expressed by a formula.

Market prices (P_m) will be used in the financial analysis and **efficiency prices (P_e)** will be used in the efficiency and referent group analyses.

1) Adjustments of market prices

If the market is perfectly competitive (no monopoly or monopsony) and does not have any distortion (such as no tax, no subsidy, no regulation, no import duty, ...), market prices of inputs and outputs may be good measures of efficiency prices. There is no any adjustment in this case.

- Such inputs are often in forms of skilled labour, local materials or transport services.
- Such outputs are often in form of foreign sales or value-added tax exemption on goods and services.

In this case, the market price and efficiency price are the same, but the analyst must also use formulas to connect them for risk analysis purposes.

1) Adjustments of market prices

If the market are not competitive (monopoly or monopsony) and have some kind of distortions (such as taxes, subsidies, regulations, import duties,...), market prices of inputs and outputs are not correct measures of efficiency prices. The analyst needs to make adjustments by using economic theory.

- Such inputs are often in forms of unskilled labour, land, imported capital goods and materials or utilities.
- Such outputs are often in form of goods and services under price regulations or subject to value-added tax/sales tax.

Notes: 1) Discount rates and exchange rates are often different between financial and efficiency analyses; 2) Tax rates, levies, and import duties should be entered separately in the project data.

2) Valuing non-market costs and benefits

For inputs and outputs used or produced by a project that are not traded, and have no market price, the analyst needs to identify and estimate them in monetary term by using appropriate non-market valuation techniques, and includes them in the project data.

Non-market inputs are often in forms of environmental damage, clean-up cost or rehabilitation cost; while non-market outputs are often in forms of public goods and services produced by public projects.

For private projects, a non-market input is sometimes linked to some item of the project through a formula. It provides a conditional communication between the project owners and the government.

3) Construction of net benefit flows

- As all efficiency prices and non-market values (if any) are parallelly entered in the project data, the analyst builds up the economic resource statement (or net benefit flow, efficiency cash flow) in the same spreadsheet with financial analyses or in a different spreadsheet.
- The format of a project's economic resource statement is quite similar to the market cash flow, but its values might be different due to different prices.
- Most alternative investment criteria such as NPV, IRR, BCR are applicable. However, the social discount rates are used in economic appraisal instead.

3) Construction of net benefit flows

Year	0	1	...	...	n
1. Benefits					
a) Gross sales/Benefits					
b) Salvage values					
2. Costs					
a) Fixed capital investment costs					
b) Changes in working capital					
c) Operating costs					
d) Non-market costs					
3. Net benefit flow	Using efficiency prices This net benefit flow is used to calculate economic NPV, IRR or BCR in order to evaluate the economic efficiency of a project. = (1) – (2)				

An illustrative example

A hypothetical project



Section 1: Project Data						
Capital cost (market price)	100	Labour Input/year	20	Domestic loan	50	5 years
Capital cost (efficiency price)	93	Market wage	3	Interest rate	5%	/year
Output flow/year	10	Shadow wage	2	Pollution	25	/year
Product price (market)	10	Other costs	10	Tax rate	20%	/year
Product price (efficiency)	11	<i>(Market price = Efficiency price)</i>		Discount rates	5%	10%

Section 2A: Market Cost-Benefit Analysis (before taxes)

Year	0	1	2	3	4	5
Gross revenues		110	110	110	110	110
Capital costs	100					
Labour costs		60	60	60	60	60
Other operating costs		10	10	10	10	10
Net cash flow	-100	40	40	40	40	40.0
NPV@5% = 73.18		NPV@10% = 51.63		IRR = 29%		

Income tax

Year	0	1	2	3	4	5
Net revenues		100	100	100	100	100
Labour costs		60	60	60	60	60
Other operating costs		10	10	10	10	10
Depreciation		25	25	25	25	25
Interest payments		2.5	2.0	1.6	1.1	0.5
Taxable income		2.5	3.0	3.4	3.9	4.5
Income tax		0.5	0.6	0.7	0.8	0.9

Section 2B: Market Cost-Benefit Analysis (after taxes)

Year	0	1	2	3	4	5
Gross revenues		110	110	110	110	110
Capital costs	100					
Labour costs		60	60	60	60	60
Other operating costs		10	10	10	10	10
Sales tax		10	10	10	10	10
Income tax		0.5	0.6	0.7	0.8	0.9
Net cash flow	-100	30	29	29	29	29.1
	NPV@5% = 26.94	NPV@10% = 11.18		IRR = 14%		

Section 3: Private Cost-Benefit Analysis

Year	0	1	2	3	4	5
Gross revenues		110	110	110	110	110
Loan	50					
Capital costs	100					
Labour costs		60	60	60	60	60
Other operating costs		10	10	10	10	10
Loan repayments		11.5	11.5	11.5	11.5	11.5
Sales tax		10	10	10	10	10
Income tax		0.5	0.6	0.7	0.8	0.9
Net cash flow	-50	18	18	18	18	17.6
	NPV@5% = 26.94	NPV@10% = 17.40		IRR = 23%		

Section 4: Efficiency Cost-Benefit Analysis						
Year	0	1	2	3	4	5
Benefits		110	110	110	110	110
Capital costs	93					
Labour costs		40	40	40	40	40
Other operating costs		10	10	10	10	10
Pollution		25	25	25	25	25
Net benefit flow	-93	35	35	35	35	35
NPV@5% = 58.07		NPV@10% = 39.22		IRR = 25%		

Section 5: Referent Group Cost-Benefit Analysis (e.g. equity owner is a foreign company)						
Year	0	1	2	3	4	5
RG aggregate net benefit flow	-43.5	17.0	17.1	17.2	17.3	17.4
RG net benefit allocation						
Domestic bank	-50.0	11.5	11.5	11.5	11.5	11.5
Government	6.5	10.5	10.6	10.7	10.8	10.9
<i>Import duty</i>	6.5					
<i>Sales taxes</i>		10.0	10.0	10.0	10.0	10.0
<i>Income taxes</i>		0.5	0.6	0.7	0.8	0.9
Labour (rent)		20.0	20.0	20.0	20.0	20.0
Community (pollution)		-25.0	-25.0	-25.0	-25.0	-25.0
Total RG net benefit flow	-43.5	17.0	17.1	17.2	17.3	17.4
Check for consistency	0.0	0.0	0.0	0.0	0.0	0.0
NPV	0%	5%	10%	15%	20%	25%
Referent Group Analysis	42.7	31.1	21.8	14.2	8.0	2.8
Market Analysis	100.0	73.2	51.6	34.1	19.6	7.6
Private Analysis	38.8	26.9	17.4	9.6	3.2	-2.1
Efficiency Analysis	81.5	58.1	39.2	23.9	11.2	0.7
Check for consistency	42.7	31.1	21.8	14.2	8.0	2.8

An illustrative example

NFG Project, Campbell & Brown (2016, pp.138-139)



Table 1: Key Variables				Project		Efficiency						Project				Efficiency Pricing			
Investment Costs		No.	Price	Cost	Price	Cost	Operating Costs		No.	Price	Cost			%	Price	Cost			
(i) Fixed Investment		(units)	(\$)		(\$)		Rent on land (Ha)		100	30	3,000	Rent op. cost		0%	0.00	0			
Farm equipment (units)		4	100,000	400,000	90,909	363,636	Fuel (litres)		2,500	0.7	1,750	Fuel tax		10%	0.64	1,591			
Vehicles (units)		3	30,000	90,000	25,000	75,000	Seeds (Kg)		250	20	5,000	Seed subsidy		25%	26.67	6,667			
Buildings (m2)		250	1,000	250,000	1,000	250,000	Fertilizers (tonnes)		3	500	1,500	Fertilizer Subsidy		30%	714.29	2,143			
TOTAL				\$740,000		\$688,636	Insecticides (litres)		3,000	30	90,000	Insecticide subsidy		20%	37.50	112,500			
(ii) Working Capital							Water (ML)		900	20	18,000	Water subsidy		40%	33.33	30,000			
Fertilizer Stocks (tons)		2	500	1,000	714.29	1,429	Spares		12	1,000	12,000	Spares -duty		15%	869.57	10,435			
Insecticide stocks (Litres)		2500	30	75,000	37.50	93,750	Casual labor (days)		100	60	6,000	Labour op. cost		20%	12.00	1,200			
Equipment spare parts (units)		10	1,000	10,000	870	8,696	Administration (/month)		12	1000	12,000	Administration (/month)				12,000			
Fuel stocks (Litres)		500	0.7	350	0.64	318	Insurance (p.a.)		1%	826,350	8,263	Insurance (p.a.)			826,349.6	8,263			
TOTAL				\$86,350.0		\$104,192	Management (/month)		12	3,000	36,000	Management			3,000	36,000			
(iii) Salvage Value		10%		\$74,000		\$68,864	Miscellaneous		1	7,700	7,700	Miscellaneous			7,700	7,700			
							TOTAL (Market prices)				\$201,213	TOTAL (Efficiency Prices)				\$228,499			
Depreciation		Life(yrs)	Amount p.a.				Revenues												
	Equipmnt.	10	40,000				Apples (tons)		100	1,000	100,000	External Costs				\$10,000			
	Vehicles	5	18,000				Peaches (tons)		90	1,250	112,500								
	Buildings	20	12,500				Pears (tons)		75	1,500	112,500	Import duties							
Financing		Amount	Interest	Life (yrs)			TOTAL				\$325,000	Equipment			10%				
Loan		\$700,000	3.5%	10					Conversion factor		1000	Vehicles			20%				
Overdraft		\$40,000	5.0%	4			Capacity Output												
Discount rate =		5.0%	10.0%	15.0%			Year	2001	2002	2003	2004+								
Tax rate on profits =		25.0%					%	25%	50%	75%	100%								

	A	B	C	D	E	F	G	L	Q	V
14	EFFICIENCY CASH FLOW									
15	Year	0	1	2	3	4	5	10	15	20
16	Benefits		81.3	162.5	243.8	325.0	325.0	325.0	325.0	393.9
17	Sales		81.3	162.5	243.8	325.0	325.0	325.0	325.0	325.0
18	Salvage value									68.9
19	Market costs	688.6	161.3	114.2	171.4	228.5	228.5	228.5	228.5	124.3
20	Fixed investment cost	688.6								
21	Working capital investment		104.2							-104.2
22	Operating costs		57.1	114.2	171.4	228.5	228.5	228.5	228.5	228.5
23	Non-market costs		0.0	10.0	10.0	10.0	10.0	10.0	10.0	10.0
24	Net benefit flows	-688.6	-80.1	38.3	62.4	86.5	86.5	86.5	86.5	259.6
25										
26	External costs to the downstream users included Market cash flow is re-calculated using efficiency prices Salvage value at efficiency price assuming it is internationally tradable									
27										
28										
29										
30										
31	Social discount rates	5%	10%	15%						
32	NPV	231.3	-135.91	-333.81						
33	IRR	7.8%								

Introduction to economic appraisal

Q&A section

